

Global extension and specializations

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Scope

Local affine-isometry charts agree on open overlaps and glue over a nonempty open connected source. Continuity and interior matching yield the convex, open-ball and equal-radius closed-ball specializations with their stated hypotheses.

4 direct Cards + 7 reused prerequisites = 11 unique Cards. This count is a selected Card closure, not a source-declaration count.

[Route reading PDF](#) · [Preserved source exploration](#)

Cards in this route

- Key result: [Mankiewicz extension on open connected domains](#)
- Related specialization: [Affine extension from equal-radius closed balls](#)

[Read this route with prerequisites](#)

Dependency-first reading route

Read selected Card prerequisites before their uses. Levels are recomputed from the selected reachability-preserving projection; omitted source helpers remain traceable in the source exploration.

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Level 0 (2 Cards)

Level 0

Symmetric lens

Names the intersection of two closed balls with a common radius.

`MathlibAnnex.symmetricLens`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Midpoint preservation on a symmetric lens](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

Center transport under bounded point-reflection symmetry

Uses a doubling argument to identify the reflection centers under a bijective isometry.

`MathlibAnnex.IsometryEquiv.map_center_of_mapsTo_pointReflection`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Midpoint preservation on a symmetric lens](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1 (1 Card)

Level 1

Midpoint preservation on a symmetric lens

Identifies the centers of two lenses without prescribing images of their foci.

`MathlibAnnex.IsometryEquiv.map_midpoint_of_symmetricLens`

Immediate Card prerequisites: [Center transport under bounded point-reflection symmetry](#) · [Symmetric lens](#)

Used by in this scope: [Midpoint preservation under lens containment](#)

[Open Declaration Card](#) [Read exact source](#)

Level 2 (1 Card)

Level 2

Midpoint preservation under lens containment

Restricts a set isometry and its inverse to the two contained lenses.

`MathlibAnnex.IsometryEquiv.map_midpoint_of_symmetricLens_subset`

Immediate Card prerequisites: [Midpoint preservation on a symmetric lens](#)

Used by in this scope: [Midpoint preservation on a quarter ball](#)

[Open Declaration Card](#) [Read exact source](#)

Level 3 (1 Card)

Level 3

Midpoint preservation on a quarter ball

Supplies both lens inclusions from matching ambient balls.

`MathlibAnnex.IsometryEquiv.map_midpoint_of_mem_ball`

Immediate Card prerequisites: [Midpoint preservation under lens containment](#)

Used by in this scope: [Affine-segment preservation on a quarter ball](#)

[Open Declaration Card](#) [Read exact source](#)

Level 4 (1 Card)

Level 4

Affine-segment preservation on a quarter ball

Passes from midpoint preservation to every real parameter in a segment.

`MathlibAnnex.IsometryEquiv.map_lineMap_of_mem_ball`

Immediate Card prerequisites: [Midpoint preservation on a quarter ball](#)

Used by in this scope: [Local affine-isometry chart on a smaller ball](#)

[Open Declaration Card](#) [Read exact source](#)

Level 5 (1 Card)

Level 5

Local affine-isometry chart on a smaller ball

Constructs one ambient surjective affine isometry from the same radial map throughout.

`MathlibAnnex.IsometryEquiv.exists_affineExtension_eqOn_ball`

Immediate Card prerequisites: [Affine-segment preservation on a quarter ball](#)

Used by in this scope: [Mankiewicz extension on open connected domains](#)

[Open Declaration Card](#) [Read exact source](#)

Level 6 (1 Card)

Level 6

Mankiewicz extension on open connected domains

Glues local affine-isometry charts using their agreement on open overlaps.

`MathlibAnnex.IsometryEquiv.existsUnique_affineExtension`

Immediate Card prerequisites: [Local affine-isometry chart on a smaller ball](#)

Used by in this scope: [Convex-set extension via ambient interiors](#) · [Affine extension from open balls](#)

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Level 7 (2 Cards)

Level 7

Affine extension from open balls

Applies the open connected theorem to two positive-radius balls.

`MathlibAnnex.IsometryEquiv.existsUnique_affineExtension_ball`

Immediate Card prerequisites: [Mankiewicz extension on open connected domains](#)

Used by in this scope: None in this selected scope

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Level 7

Convex-set extension via ambient interiors

Extends agreement from a convex interior to all source points by continuity.

`MathlibAnnex.IsometryEquiv.existsUnique_affineExtension_of_convex`

Immediate Card prerequisites: [Mankiewicz extension on open connected domains](#)

Used by in this scope: [Affine extension from equal-radius closed balls](#)

[Open Declaration Card](#) [Read exact source](#)

Level 8 (1 Card)

Level 8

Affine extension from equal-radius closed balls

Recovers the centers, matches the interiors, and then extends across the boundary.

`MathlibAnnex.IsometryEquiv.existsUnique_affineExtension_closedBall`

Immediate Card prerequisites: [Convex-set extension via ambient interiors](#)

Used by in this scope: None in this selected scope

[Open Declaration Card](#) [Read exact source](#)